

Synchronized Serves II

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The serve is easily the most complicated stroke in tennis and one area that seems to provide some major differences between the men's and women's game. The question needs to be asked as to why we notice a difference in serve speed, spin and consistency at the junior, collegiate and professional level between male and female players. Is it a strength issue, a technique issue and priority of training issue? All these components may play a role, but to be successful at any level having a technically proficient service motion is vital to improved results. In last issue (Vol. 11 No. 1), we analyzed two male servers in the professional game, Andy Roddick and Fernando Verdasco. In this issue we are focusing on two of the best servers in the women's game – Sam Stosur and Justine Henin. The complexity of the serve and coordination of the movements require the effective utilization of the ground to push against and help transfer the forces developed from the ground up through the kinetic chain and out into the ball. If any of the links in the chain are not synchronized effectively the outcome of the serve will not be optimal. It is generally noticed that the synchronization of these links are not as consistent in female servers and the purpose of this article is to highlight eight major areas to focus on when analyzing or developing an effective high performance serve.

The serve can be broken down into 3 Phases and 8 Stages (including the start and finish).

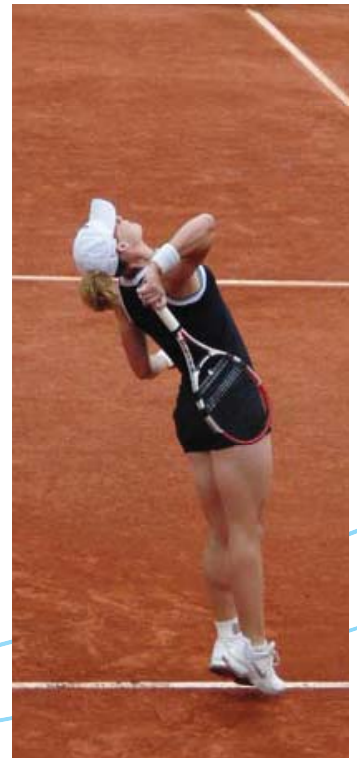
Phase	Stage
Preparation	1. Start
	2. Release
	3. Loading
	4. Cocking
Acceleration	5. Acceleration
	6. Contact
Follow-Through	7. Deceleration
	8. Finish

PREPARATION

1. **START** - Photo 1 highlights the starting position of the serve and it is clear that both players are relaxed and just about to start the weight transfer (i.e. rocking back) to initiate muscle contractions in the back leg. At this point in the service motion, most good servers have a clear picture of where they are going to hit the serve (i.e. wide, into the body or a t serve) and what type of spin they will be imparting on the ball. Having a repeatable pre-service routine is a hallmark of good servers and something that needs to be developed in servers at all levels.
2. **RELEASE** - Photo 2 highlights the release of the ball from the non-dominant hand (left hand for both

servers). As the ball toss is one area that many players struggle to perform consistently (especially under pressure), it needs to be simple to perform and easy to replicate. Notice that both players have released the ball, after their tossing arm has elevated from the inside of the front leg, in-line with the baseline. This ball release motion is preferable to a straight on (perpendicular with the baseline) ball-toss release as it allows for greater rotational loading (i.e. hip and shoulder turn). This is one area that both Stosur and Henin do extremely well and is something that is a major factor in effectively synchronizing the following six components of the serve. It is also important to note that both players release at the same relative height - when their tossing arm is just above the eye level.

3. **LOADING** – Photo 3 showcases the loading position for both players. Efficient loading mechanics are displayed, though Henin uses a platform (or foot back) technique, whereas Stosur uses a pinpoint (or foot up) technique. It is important to highlight the use of the back leg involvement during this loading phase. Both players do a good job of pushing hard into the ground during this loading stage which develops high levels of potential energy (stored energy), which will be utilized as kinetic energy (usable energy) during the acceleration phase of the serve. This back leg involvement during the loading stage is one area that both players execute well, and many other female servers at all levels of the game sometimes struggle to effectively load. Both athletes have full non-dominant arm extension, and each player's front hip is pushing into the court - which also aids in the storage of greater amounts of energy that can be converted during the remainder of the serve.
4. **COCKING** – Photo 4 highlights how similar effective servers should be at this point in the service motion. Although both players use different starting mechanics in the upper and lower body, at the cocking stage of the service motion they have nearly identical positions. Notice the racket head is pointing to the ground (requiring phenomenal external rotation at the shoulder). A line can be drawn from the dominant to the non-dominant shoulder (termed "shoulder-over-shoulder" position) and also from the back hip to the front hip ("hip-over-hip" position). This position is one of the biggest differences between effective and ineffective servers and less female servers (at all levels of the game) are consistently able to successfully navigate to this position compared to the male counterparts. It is interesting to note the Stosur is able to achieve a greater range of motion at this point and is the reason she is able to generate so much spin on her serve. Players who do not reach this position in the service motion, usually suffer from a poor ball toss and/or opening up the back hip/shoulder too early in the motion. Lack of shoulder flexibility also can contribute to players not being able to reach a full effective cocking stage.

JUSTINE HENIN SERVE SEQUENCE**1****2****3****4****SAMANTHA STOSUR SERVE SEQUENCE**



5



6



7



8



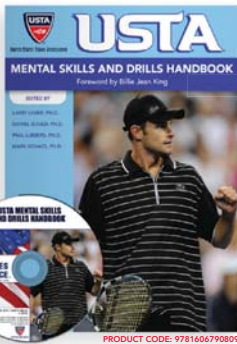
ACCELERATION

5. **ACCELERATION** – Photo 5 is the release of energy which has been stored in the previous stages of the serve. This rapid acceleration of both the lower and upper body is the result of effective summation of forces from the ground up through the kinetic chain which results in ball contact (photo 6). Notice that both players have maintained a stable body position highlighting the shoulder-over-shoulder and hip-over-hip position, while also keeping their head up and eyes looking up. Too often less effective servers will drop their head early in the acceleration stage and this can limit velocity, consistency and can also increase the risk of injury.
6. **CONTACT** - Photo 6 demonstrates contact position for both Stosur and Henin. These contact points are a direct result of the positions that both players were able to progress through during the previous five stages of the serve. It is interesting to note that although the contact points are less extended when compared to many of the male servers, Stosur and Henin are in effective contact positions with the upper body fully extended and non-dominant hand arm tucked into the waist. Due to an effective leg drive both players are substantially off the ground at ball contact.

FOLLOW-THROUGH

7. **DECELERATION** – Photo 7 requires a transition of energy from exploding through the ball to decelerating the body (specifically the upper body) after impact. These photos show that both players have successfully made contact with the ball and have finished the follow-through. This included vigorous long-axis rotation (which is a combination of internal shoulder rotation and forearm/wrist pronation). All good servers perform this component. It is also important to observe that both players have maintained good upper body position, keeping their head and chest up throughout this stage of the serving motion. Most ineffective servers at this point in the motion have collapsed at the core, resulting in the head and chest pulling down which negatively impacts both velocity and consistency of the serve.
8. **FINISH** – Photo 8 demonstrates a great finish (landing) position for the serve. This position requires great lower body deceleration to help cushion the landing on the front leg (left leg). Both players have very similar landing mechanics and it is clear that their head is looking forward at the opponent which allows them to effectively react to the return of serve. The front leg has a slight bend at landing which works as a shock absorber to lessen the impact on the knee, hip and lower back. The back leg kicks up and back, which is in direct response to the front leg landing on the ground.


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